

Lab 4

Step 1: Archivelog mode

- Show your archiving status then enable the archiving mode if not enabled and show your archiving status again.

```
SQL> archive log list;
Database log mode                No Archive Mode
Automatic archival               Disabled
Archive destination              USE_DB_RECOVERY_FILE_DEST
Oldest online log sequence      5
Current log sequence            7
SQL> shutdown immediate
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL> startup mount
ORACLE instance started.

Total System Global Area 2147481656 bytes
Fixed Size                  8898616 bytes
Variable Size              486539264 bytes
Database Buffers           1644167168 bytes
Redo Buffers                7876608 bytes
Database mounted.
SQL> alter database archivelog
  2 ;

Database altered.

SQL> alter database open;

Database altered.

SQL> archive log list;
Database log mode                Archive Mode
Automatic archival               Enabled
Archive destination              USE_DB_RECOVERY_FILE_DEST
Oldest online log sequence      5
Next log sequence to archive    7
Current log sequence            7
SQL>
```

Step 2: Alert log file

- Identify the location of the alert log file.

```
SQL> show parameter back
```

NAME	TYPE	VALUE
background_core_dump	string	partial
background_dump_dest	string	/u01/app/oracle/product/19.0/d bhome_1/rdbms/log
backup_tape_io_slaves	boolean	FALSE
db_flashback_retention_target	integer	1440
fast_start_parallel_rollback	string	LOW
rollback_segments	string	
transactions_per_rollback_segment	integer	5

```
SQL> select * from v$diag_info;
```

INST_ID	NAME	VALUE
1	Diag Enabled	TRUE
1	ADR Base	/u01/app/oracle
1	ADR Home	/u01/app/oracle/diag/rdbms/sadb/SADB
1	Diag Trace	/u01/app/oracle/diag/rdbms/sadb/SADB/trace
1	Diag Alert	/u01/app/oracle/diag/rdbms/sadb/SADB/alert
1	Diag Incident	/u01/app/oracle/diag/rdbms/sadb/SADB/incident
1	Diag Cdump	/u01/app/oracle/diag/rdbms/sadb/SADB/cdump
1	Health Monitor	/u01/app/oracle/diag/rdbms/sadb/SADB/hm
1	Default Trace File	/u01/app/oracle/diag/rdbms/sadb/SADB/trace/SADB_ora_5661.trc
1	Active Problem Count	0
1	Active Incident Count	0
1	ORACLE_HOME	/u01/app/oracle/product/19.0/dbhome_1

12 rows selected.

```
[oracle@oraclevm ~]$ adrci
```

ADRCI: Release 19.0.0.0.0 - Production on Fri Jun 12 07:15:57 2026

Copyright (c) 1982, 2019, Oracle and/or its affiliates. All rights reserved.

ADR base = "/u01/app/oracle"

```
adrci> homes
```

DIA-48415: Syntax error found in string [homes] at column [5]

```
adrci> home
```

DIA-48415: Syntax error found in string [home] at column [4]

```
adrci> show homes;
```

ADR Homes:

diag/rdbms/sadb/SADB

diag/tnslsnr/oraclevm/listener

```
adrci>
```

Step 3: Monitor the alert log

- Shutdown "SADB" with the most safe and fast method. and monitor the alert log

```
tail -100 /u01/app/oracle/diag/rdbms/sadb/SADB/alert/log.xml
```

```
<msg time='2026-06-12T08:55:48.353-07:00' org_id='oracle' comp_id='rdbms'  
type='UNKNOWN' level='16' host_id='oraclevm'  
host_addr='127.0.0.1' pid='33111'>  
<txt>.... (PID:33111): Archival disabled due to shutdown: 1089  
</txt>  
</msg>  
<msg time='2026-06-12T08:55:48.353-07:00' org_id='oracle' comp_id='rdbms'  
msg_id='krso_proc_stop:2494:825284245' type='NOTIFICATION' group='shutdown'  
level='16' host_id='oraclevm' host_addr='127.0.0.1'  
pid='33111'>  
<txt>Shutting down archive processes  
</txt>  
</msg>  
<msg time='2026-06-12T08:55:48.353-07:00' org_id='oracle' comp_id='rdbms'  
msg_id='krso_proc_stop:2534:1369042098' type='NOTIFICATION' group='shutdown'  
level='16' host_id='oraclevm' host_addr='127.0.0.1'  
pid='33111'>  
<txt>Archiving is disabled  
</txt>  
</msg>  
<msg time='2026-06-12T08:55:48.354-07:00' org_id='oracle' comp_id='rdbms'  
type='UNKNOWN' level='16' host_id='oraclevm'  
host_addr='127.0.0.1' pid='25527'>  
<txt>Stopping background process VKTM  
</txt>  
</msg>  
<msg time='2026-06-12T08:55:48.410-07:00' org_id='oracle' comp_id='rdbms'  
type='UNKNOWN' level='16' host_id='oraclevm'  
host_addr='127.0.0.1' pid='33111'>  
<txt>JIT: pid 33111 requesting stop  
</txt>  
</msg>  
<msg time='2026-06-12T08:55:58.671-07:00' org_id='oracle' comp_id='rdbms'  
msg_id='opistp_real:2303:4046551520' type='NOTIFICATION' group='shutdown'  
level='16' host_id='oraclevm' host_addr='127.0.0.1'  
pid='33111'>  
<txt>Instance shutdown complete (OS id: 33111)  
</txt>  
</msg>  
[oracle@oraclevm ~]$
```

Step 4:

- Startup "SADB" step by step. and monitor the alert log

```
SQL> startup nomount
ORACLE instance started.

Total System Global Area 2147481656 bytes
Fixed Size                  8898616 bytes
Variable Size              486539264 bytes
Database Buffers          1644167168 bytes
Redo Buffers               7876608 bytes
SQL> alter database mount
    2 ;

Database altered.

SQL> alter database open
    2 ;

Database altered.

SQL> █
```

at startup nomount

```
<msg time='2026-06-12T08:57:45.725-07:00' org_id='oracle' comp_id='rdbms'
type='UNKNOWN' level='16' host_id='oraclevm'
host_addr='127.0.0.1' pid='50853'>
<txt>starting up 1 shared server(s) ...
</txt>
</msg>
<msg time='2026-06-12T08:57:45.734-07:00' org_id='oracle' comp_id='rdbms'
msg_id='ksbs1p_int:6681:1349786505' type='NOTIFICATION' group='process start'
level='16' host_id='oraclevm' host_addr='127.0.0.1'
pid='50853'>
<txt>Starting background process TMON
</txt>
</msg>
<msg time='2026-06-12T08:57:45.742-07:00' org_id='oracle' comp_id='rdbms'
msg_id='ksbrdp:9853:1214121915' type='NOTIFICATION' group='process start'
level='16' host_id='oraclevm' host_addr='127.0.0.1'
pid='51672'>
<txt>TMON started with pid=35, OS id=51672
</txt>
</msg>
<msg time='2026-06-12T08:57:45.743-07:00' org_id='oracle' comp_id='rdbms'
type='UNKNOWN' level='16' host_id='oraclevm'
host_addr='127.0.0.1' pid='50853'>
<txt>Setting CPU count to 2
</txt>
</msg>
<msg time='2026-06-12T08:57:45.743-07:00' org_id='oracle' comp_id='rdbms'
msg_id='ksu_setup_oracle_base:35408:2787919602' type='NOTIFICATION' group='startup'
level='16' host_id='oraclevm' host_addr='127.0.0.1'
pid='50853'>
<txt>ORACLE_BASE from environment = /u01/app/oracle
</txt>
</msg>
<msg time='2026-06-12T08:57:48.510-07:00' org_id='oracle' comp_id='rdbms'
type='UNKNOWN' level='16' host_id='oraclevm'
host_addr='127.0.0.1' pid='51627'>
<txt>Using default pga_aggregate_limit of 2048 MB
</txt>
</msg>
[oracle@oraclevm ~]$ █
```

at alter database mount

```

<msg time='2026-06-12T08:59:10.715-07:00' org_id='oracle' comp_id='rdbms'
msg_id='opiexe:3766:4222364190' type='NOTIFICATION' group='admin_ddl'
level='16' host_id='oraclevm' host_addr='127.0.0.1'
module='sqlplus@oraclevm (TNS V1-V3)' pid='51674'>
<txt>alter database mount
</txt>
</msg>
<msg time='2026-06-12T08:59:20.735-07:00' org_id='oracle' comp_id='rdbms'
type='UNKNOWN' level='16' host_id='oraclevm'
host_addr='127.0.0.1' module='sqlplus@oraclevm (TNS V1-V3)' pid='51674'>
<txt>... (PID:51674): Redo network throttle feature is disabled at mount time
</txt>
</msg>
<msg time='2026-06-12T08:59:20.749-07:00' org_id='oracle' comp_id='rdbms'
type='UNKNOWN' level='16' host_id='oraclevm'
host_addr='127.0.0.1' pid='51639'>
<txt>Successful mount of redo thread 1, with mount id 2092197092
</txt>
</msg>
<msg time='2026-06-12T08:59:20.749-07:00' org_id='oracle' comp_id='rdbms'
type='UNKNOWN' level='16' host_id='oraclevm'
host_addr='127.0.0.1' module='sqlplus@oraclevm (TNS V1-V3)' pid='51674'>
<txt>Database mounted in Exclusive Mode
</txt>
</msg>
<msg time='2026-06-12T08:59:20.750-07:00' org_id='oracle' comp_id='rdbms'
type='UNKNOWN' level='16' host_id='oraclevm'
host_addr='127.0.0.1' module='sqlplus@oraclevm (TNS V1-V3)' pid='51674'>
<txt>Lost write protection disabled
</txt>
</msg>
<msg time='2026-06-12T08:59:20.750-07:00' org_id='oracle' comp_id='rdbms'
type='UNKNOWN' level='16' host_id='oraclevm'
host_addr='127.0.0.1' module='sqlplus@oraclevm (TNS V1-V3)' pid='51674'>
<txt>... (PID:51674): Using STANDBY_ARCHIVE_DEST parameter default value as USE_DB_RECOVERY_FILE_D
EST [krtd.c:18157]
</txt>
</msg>
<msg time='2026-06-12T08:59:20.788-07:00' org_id='oracle' comp_id='rdbms'
msg_id='opiexe:3870:2802784106' type='NOTIFICATION' group='admin_ddl'

```

at alter database open:

```

<msg time='2026-06-12T09:00:58.421-07:00' org_id='oracle' comp_id='rdbms'
type='UNKNOWN' level='16' host_id='oraclevm'
host_addr='127.0.0.1' pid='51663'>
<txt>29465047
</txt>
</msg>
<msg time='2026-06-12T09:00:58.421-07:00' org_id='oracle' comp_id='rdbms'
type='UNKNOWN' level='16' host_id='oraclevm'
host_addr='127.0.0.1' pid='51663'>
<txt>=====
</txt>
</msg>
<msg time='2026-06-12T09:00:58.568-07:00' org_id='oracle' comp_id='rdbms'
type='UNKNOWN' level='16' host_id='oraclevm'
host_addr='127.0.0.1' module='MMON_SLAVE' pid='51694'>
<txt>db_recovery_file_dest_size of 8256 MB is 0.00% used. This is a
</txt>
</msg>
<msg time='2026-06-12T09:00:58.568-07:00' org_id='oracle' comp_id='rdbms'
type='UNKNOWN' level='16' host_id='oraclevm'
host_addr='127.0.0.1' module='MMON_SLAVE' pid='51694'>
<txt>user-specified limit on the amount of space that will be used by this
</txt>
</msg>
<msg time='2026-06-12T09:00:58.568-07:00' org_id='oracle' comp_id='rdbms'
type='UNKNOWN' level='16' host_id='oraclevm'
host_addr='127.0.0.1' module='MMON_SLAVE' pid='51694'>
<txt>database for recovery-related files, and does not reflect the amount of
</txt>
</msg>
<msg time='2026-06-12T09:00:58.568-07:00' org_id='oracle' comp_id='rdbms'
type='UNKNOWN' level='16' host_id='oraclevm'
host_addr='127.0.0.1' module='MMON_SLAVE' pid='51694'>
<txt>space available in the underlying filesystem or ASM diskgroup.
</txt>
</msg>
[oracle@oraclevm ~]$ █

```

Step 5: Create a user

- Create user "myuser" and make "iti_data" tablespace his default tablespace with unlimited quota on it.

```

SQL> create user myuser identified by 123;

User created.

SQL> alter user myuser default tablespace iti_data;

User altered.

SQL> alter user myuser qouta unlimited on iti_data;
alter user myuser qouta unlimited on iti_data
      *
ERROR at line 1:
ORA-00922: missing or invalid option

SQL> alter user myuser quota unlimited on iti_data;

User altered.

```

Step 6: Connection trials

- try to connect with the new created user.

```
SQL> conn myuser/123
ERROR:
ORA-01045: user MYUSER lacks CREATE SESSION privilege; logon denied

Warning: You are no longer connected to ORACLE.
SQL>
```

Step 7: Giving user privileges

- give the new user connect, resource and create view privileges.

```
SQL> conn sys/123 as sysdba
Connected.
SQL> grant connect , resource , create view to myuser;

Grant succeeded.
```

Step 8: Monitor the alert log again

- shutdown "SADB" with the most safe and fast method. and monitor the alert log

```
<msg time='2026-06-12T07:57:02.055-07:00' org_id='oracle' comp_id='rdbms'
type='UNKNOWN' level='16' host_id='oraclevm'
host_addr='127.0.0.1' pid='25581'>
<txt>=====
</txt>
</msg>
<msg time='2026-06-12T07:57:02.242-07:00' org_id='oracle' comp_id='rdbms'
type='UNKNOWN' level='16' host_id='oraclevm'
host_addr='127.0.0.1' module='MMON_SLAVE' pid='25613'>
<txt>db_recovery_file_dest_size of 8256 MB is 0.00% used. This is a
</txt>
</msg>
<msg time='2026-06-12T07:57:02.243-07:00' org_id='oracle' comp_id='rdbms'
type='UNKNOWN' level='16' host_id='oraclevm'
host_addr='127.0.0.1' module='MMON_SLAVE' pid='25613'>
<txt>user-specified limit on the amount of space that will be used by this
</txt>
</msg>
<msg time='2026-06-12T07:57:02.243-07:00' org_id='oracle' comp_id='rdbms'
type='UNKNOWN' level='16' host_id='oraclevm'
host_addr='127.0.0.1' module='MMON_SLAVE' pid='25613'>
<txt>database for recovery-related files, and does not reflect the amount of
</txt>
</msg>
<msg time='2026-06-12T07:57:02.243-07:00' org_id='oracle' comp_id='rdbms'
type='UNKNOWN' level='16' host_id='oraclevm'
host_addr='127.0.0.1' module='MMON_SLAVE' pid='25613'>
<txt>space available in the underlying filesystem or ASM diskgroup.
</txt>
</msg>
[oracle@oraclevm ~]$ tail -100 /u01/app/oracle/diag/rdbms/sadb/SADB/alert/log.xml
host_addr='127.0.0.1' pid='25581'>
<txt>29111631,29112455,29117337,29123444,29125708,29125786,29129476,29131772,
```

Step 9: Unlock the hr user if locked.


```
SQL> alter user hr account unlock;
alter user hr account unlock
      *
ERROR at line 1:
ORA-01918: user 'HR' does not exist

SQL>
```

As far as i can remember, we didn't install the hr schema while installation.

- checking status of myuser:

```
SQL> select username , account_status from dba_users where username = 'MYUSER';

USERNAME
ACCOUNT_STATUS
-----
MYUSER
OPEN
```

Step 10: Create ROLE

- create role "myrole" including "create session" privilege, create view. --> to know what privileges in a role, query on dba_sys_privs view.

```
SQL> create role myrole;

Role created.

SQL> grant create session, create view to myrole;

Grant succeeded.

SQL> select * from dba_sys_privs where grantee = 'MYROLE';

GRANTEE PRIVILEGE                                ADM COM INH
-----
MYROLE CREATE VIEW                                NO  NO  NO
MYROLE CREATE SESSION                            NO  NO  NO
```

Step 11: Create PROFILE

- create profile "myprofile" which restrict the password age of the user to be 5 days, number of failed logins to three times and restrict its connect time.

```
SQL> create profile myprofile limit password_life_time 5 failed_login_attempts 3 connect_time 60;

Profile created.
```

Step 12:

- create an OS user with name "iti" and make him login to the database using "sqlplus /"


```
[root@oraclevm ~]# useradd -s /bin/bash -m iti
[root@oraclevm ~]# passwd iti
Changing password for user iti.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@oraclevm ~]#
```

```
[iti@oraclevm ~]$ sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Fri Jun 12 09:20:23 2026
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

ERROR:
ORA-01017: invalid username/password; logon denied

Enter user-name:
```

Step 13:

- make your personal Operating system user to connect locally without authentication as sysdba.

solution was to add iti to dba group

```
[root@oraclevm ~]# usermod -aG dba iti
[root@oraclevm ~]#
```

```
[iti@oraclevm ~]$ sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Fri Jun 12 09:22:05 2026
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL>
```

Step 14:

- grant sysdba privilege to your created user "myuser" and then show all users that have sysdba privilege.

```
SQL> grant sysdba to myuser;  
Grant succeeded.
```

```
SQL> select username from v$pwfile_users where lower(sysdba) = 'true';  
USERNAME  
-----  
-----  
SYS  
MYUSER
```